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Cryptographic Providers

In Microsoft SQL Server

Cryptographic Providers in SQL Server

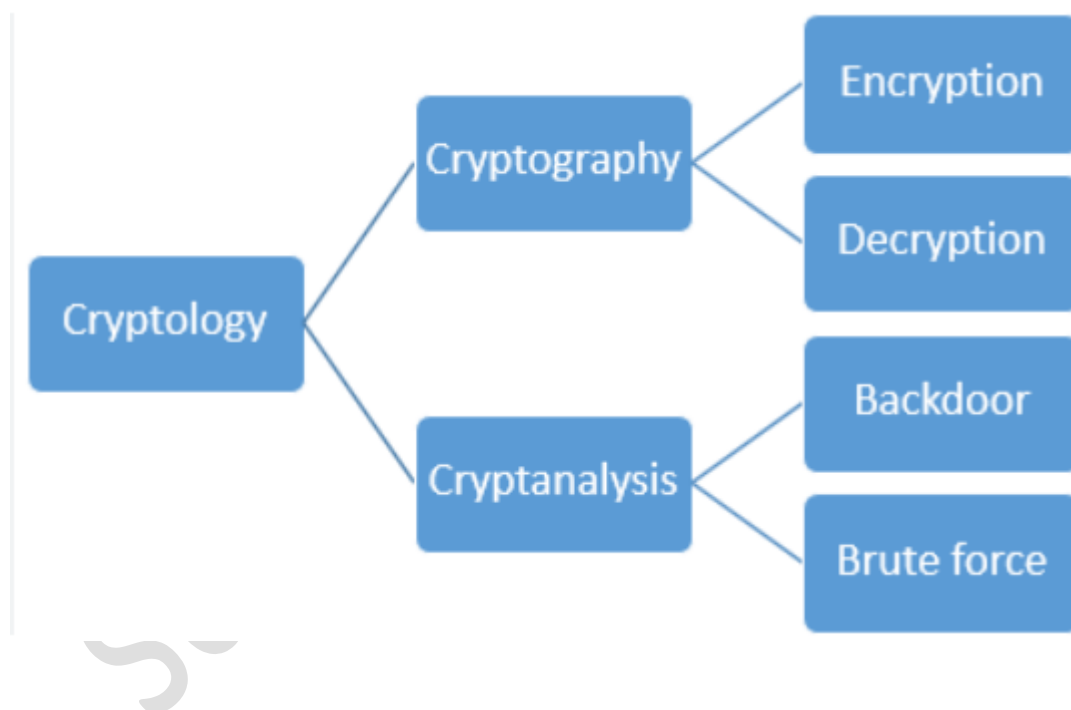
Cryptographic Providers in **SQL Server** are **third-party or built-in modules** that enable **encryption and decryption** for securing sensitive data. They work with **Extensible Key Management (EKM)** and **Transparent Data Encryption (TDE)** to protect data at rest.

Why Use Cryptographic Providers?

- ✓ **Data Security** – Protects **sensitive data** from unauthorized access.
- ✓ **Compliance** – Helps meet **regulatory standards** (e.g., GDPR, HIPAA, PCI DSS).
- ✓ **Key Management** – Allows **external key storage** using Hardware Security Modules (HSMs).
- ✓ **Encryption Support** – Works with **TDE, Always Encrypted, and Symmetric/Asymmetric encryption**.

Types of Cryptographic Providers

1. **Microsoft Software Key Storage Provider** – Default built-in provider for managing encryption keys.
2. **Microsoft Smart Card Key Storage Provider** – Uses smart cards for key storage.
3. **Third-Party EKM Providers** – External vendors like **Thales, SafeNet** provide HSM-based encryption.



How to Work with Cryptographic Providers in SQL Server?

1. Creating a Cryptographic Provider

```
CREATE CRYPTOGRAPHIC PROVIDER SaiReddProvider
```

```
FROM FILE = 'C:\Program Files\CustomEKM.dll';--replace with actual dll
```

Known Issues: Msg 33053,

Extensible Key Management is disabled or not supported on this edition of SQL Server.

Use sp_configure 'EKM provider enabled' to enable it.

```
SELECT SERVERPROPERTY('Edition');
sp_configure 'show advanced options', 1;
GO
RECONFIGURE;
GO
sp_configure 'EKM provider enabled', 1;
GO
RECONFIGURE;
GO
```

2. Viewing Existing Providers

```
SELECT * FROM sys.cryptographic_providers;
```

3. Altering a Cryptographic Provider

```
ALTER CRYPTOGRAPHIC PROVIDER SaiReddProvider
ENABLE;
```

4. Deleting a Cryptographic Provider

```
DROP CRYPTOGRAPHIC PROVIDER SaiReddProvider ;
```

When to Use Cryptographic Providers?

- ✦ Encrypting databases with Transparent Data Encryption (TDE).
- ✦ Storing encryption keys securely in an HSM or external provider.
- ✦ Implementing Always Encrypted for protecting sensitive columns.
- ✦ Meeting compliance standards in finance, healthcare, and government sectors.

Advantages & Disadvantages of Cryptographic Providers

Advantages:

- ✓ Enhances data security.
- ✓ Supports external key management (HSMs).
- ✓ Helps with regulatory compliance.
- ✓ Works with TDE and Always Encrypted.

Disadvantages:

- ⚠ Performance Overhead – Encryption may impact query speed.
- ⚠ Complex Setup – Requires configuration and key management.
- ⚠ External Dependency – HSM-based providers depend on third-party vendors.

🔒 SQL Server Cryptographic Providers – Why & When to Use?